

Review

Algae: The Reservoir of Bioethanol

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Abstract: Overuse of non-renewable fossil fuels due to the population explosion urges us to focus on renewable fuels such as bioethanol. It is a well-known fact that ethanol is useful as a blending product with common fuels such as petrol and diesel. This reduces the cost besides bringing down environmental pollution. Apart from chemical methods, bioethanol is generated from photosynthetic plants including algae, plant-based products, microbial organisms and their waste. Specifically, the production of ethanol from microalgal sources has been an attractive method in recent days. The reason behind using microalgal species is their simple structure with photosynthetic ability. In contrast, certain algal species often go disused in some regions. Hence, the production of ethanol from algal sources is one of the best waste management practices. Moreover, it is easy to improve the biomass in microalgal species by altering the physicochemical conditions such as light, pH, temperature, external supply of nutrients, vitamins, nano-sized particles, gene alterations etc., which will enhance ethanol production. In this review, the methods used for ethanol production are discussed. In addition, the factors involved in algal growth and ethanol production are emphasized. Overall, this review focuses on ethanol production from various algal species. This information will be useful for industrial-level production of ethanol and future renewable energy research.

Keywords: algae; photosynthesis; biomass; fermentation; bioethanol; enhancement



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1. Introduction

Current lifestyles undoubtedly make us dependent on a continuous supply of energy. In addition, the population explosion and their necessities have been progressing toward a conflict with energy obligations in recent times [1]. In contrast, regular consumption of fossil fuels depletes a major portion of non-renewable energy sources. According to the world energy outlook 2015, fossil fuels such as coal, petrol products and natural gas are being used predominantly for energy requirements in various sectors, providing more than 80% of the world's energy between 2013 and 2035 [2]. Specifically, energy demands across the industrial and transport sectors are high compared to other sectors [3,4]. The gap between demand and supply of fossil fuels is increasing continuously, thereby raising

prices in recent days, which, in turn, affects the economy of the country [5,6]. Hence, there is a need to look into alternate sources of energy that are cost-effective, causes no threat to the environment and at the same time are renewable in nature. Though solar energy, hydroenergy and wind energy are in commercial practice now, more efforts are necessary for the industrial practice of biofuels [7].

Biomass-based energy is another important renewable source. These biomass-based fuels are mainly produced from biological organic matter derived from plants, animals, microbial organisms and their wastes including dead organisms [8]. Initially, biofuel research was mainly focused on advanced plants. The quantity and quality of biomass is the key factor for biofuel production from algae [9,10]. In addition, biomass and its chemical composition differ amongst different species and cultivars/subspecies. Biomass consists of a high percentage of carbohydrates, which is a source of ethanol production, whereas high fatty acids/lipids generate diesel [11,12]. In contrast, the excessive exploitation of land resources for biomass has led to forest, land and soil degradation [9]. Hence, using limited space in a bioreactor system with algal species is one of the best ways to generate biofuels. The production of biofuels in algae involves contemporary CO₂ fixation, such as that which occurs in advanced plants in the process of photosynthesis [8]. Also, it is important to understand the influence of various physicochemical properties on biofuel production. Most of the researchers prefer algae over other organisms because they belong to a group of primitive plants with efficient photosynthetic capacity and produce greater amounts of biomass in the form of carbohydrates [13]. The life cycles of most algae are simple and are responsible for 50 percent of the planet's atmospheric carbon fixation. In certain countries, these algal species are ignored, not utilized much and are often considered as waste. Hence, generation of fuels from algal species is a kind of waste management practice [14].

Ethanol or bioethanol (C₂H₅OH) is often blended with petroleum products and used for transportation purposes, which reduces the cost as well as pollution. Generally, ethanol burns and releases CO₂ and water, which is not countable when compared with other types of pollutants [15]. Ethanol fuel blends are widely sold in the United States of America. The most common blend is 90% petrol and 10% ethanol, i.e., E10 [16]. The new Biofuel Policy-2018 fixed a target to achieve 20 percent ethanol blending with petrol by 2030. Few studies have reported that only flexible fuel vehicles can run on up to 85% ethanol and 15% petrol blends [17]. Apart from energy usage, ethanol is used as a beverage and is also useful for the preparation of certain medicines and syrups. Therefore, bioethanol tends to provide a higher degree of national energy security in an eco-friendly and sustainable manner by adding conventional energy resources, decreasing dependence on imported fossil fuels and meeting energy needs [18]. With this background, this review intends to summarize the progress in ethanol production methods from algal species while keeping its excellent prospects in view. This review covers the influence of various factors on algal growth and ethanol production levels. In addition, various algal species used for bioethanol production are highlighted. This knowledge may be helpful in bridging the gap between demand and supply of fossil fuels and is also useful for the commercial production of bioethanol from algal species.

2. Production of Ethanol

Ethanol is one of the environmentally friendly fuels that can be produced by both chemical and biological processes, as mentioned in Figure 1. Chemically, ethanol can be manufactured through the transformation of ethylene with steam [19]. Ethanol formation takes place with the liberation of heat energy, which is considered exothermic. In normal conditions, the equilibrium will be shifted and a small amount of ethanol will be obtained. Thus, to increase the yield of ethanol significantly, reaction is carried out at 300 °C and at 60–70 atmospheric pressure using a phosphoric acid (H₃PO₄) catalyst [20,21]. A number of physicochemical factors are involved in the enhancement of ethanol production. Specifically, there are several reports available regarding the utility of various metal phosphates (Ge, Zr, Ti, Sn etc.) and phosphoric-acid-impregnated metal phosphates at high pressures for the

increased production of ethanol. The rate of ethanol production with different catalysts was described in detail by Hidzir et al. [19]. In addition, efficient catalytic conversion of syngas to ethanol has been developed for commercial practice [22].

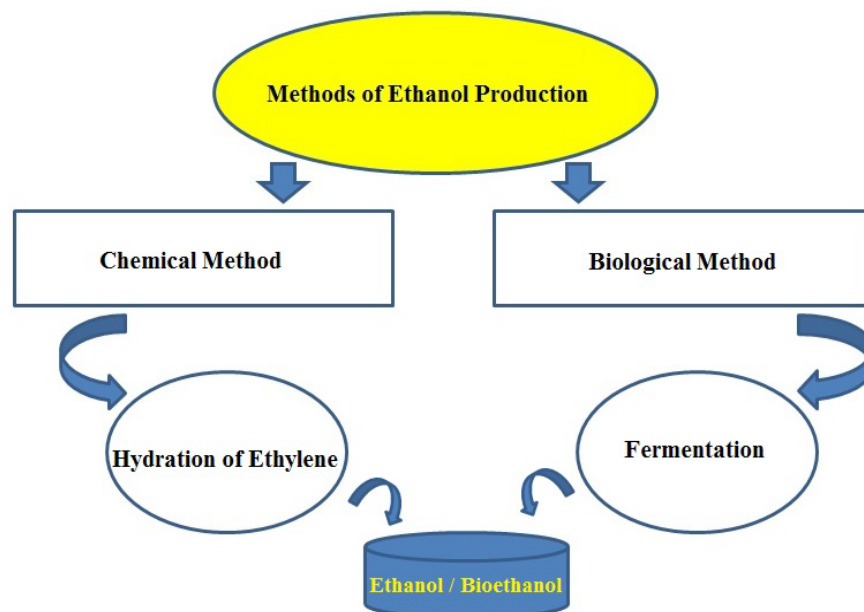


Figure 1. Schematic representation of ethanol production methods.

Biologically, fermentation is the most widely used traditional method of alcohol production. Fermentation is a process in which fermentable sugars are converted to ethanol by microorganisms [23,24]. It is a simple process with less consumption of energy and became popular in ethanol production from the biomass of photosynthetic organisms. Few organisms such as yeast, bacteria and fungi have been used in the process of fermentation, which include *Saccharomyces cerevisiae*, *Zymomonas mobilis*, *Clostridium pasteurianum* etc. [25–28]. Bioethanol yield from algae depends on factors such as biomass content, pH, temperature, alcohol tolerance, resistance to inhibitors, growth rate, osmotic tolerance and genetic stability [29]. In general, the fermentation process is carried out using two basic types of processes, i.e., simultaneous saccharification and fermentation (SSF) and separate hydrolysis and fermentation (SHF) [30]. In the SSF method, pretreatment, enzymatic hydrolysis and fermentation of algal biomass were carried out in one unit; in the SHF process, enzymatic hydrolysis is performed separately, followed by fermentation [29,31,32]. Both processes have advantages and disadvantages but the SHF process takes less residence time, is cost-effective and only needs simple equipment; hence, it can be encouraged for large-scale applications [33]. Moreover, simultaneous saccharification and co-fermentation (SSCF) and separate hydrolysis and co-fermentation (SHCF) are the extension methods of SSF and SHF, respectively. Certain strains of yeast cannot convert carbohydrates such as pentose under moderate conditions, which leads to impurity in the biomass and decreases ethanol yield [24]. The generation of recombinant strains may be helpful to convert residues such as pentose to ethanol.

Most of the world's bioethanol is produced by fermentation, initially using photosynthetic crops such as corn, sugarcane, maize, sugar beet and rice and later using non-edible crops, algae, diatoms, certain bacteria and their waste [34,35]. For instance, ethanol is produced from corn kernels through milling, liquefaction, saccharification, fermentation, distillation, dehydration and denaturation. Further, most of the municipal waste can also be used as feedstock for ethanol production [36]. Fermentation is a complex series of reactions that convert carbohydrates mainly sugars, starch, cellulose, hemicellulose and lignocellulosic materials into ethanol and CO₂ [18]. It works best at temperatures in the

range of 25 °C–37 °C in the absence of oxygen and will produce aqueous solutions of upto 14% ethanol. Higher alcohol contents require further distillation [23].

3. Biological Sources Used for Ethanol Production

Bioenergy is another form of natural source that is initially produced from plants. Normally, plant materials such as leaves, wood, seeds and whole plants are useful for fuel generation. Biofuels have been classified into four generations (Table 1) by several researchers [37]. The first-generation biofuels are generated from edible crops such as corn, sugarcane, maize, cassava etc. Due to reasons such as food security, investigations continued using non-food crops, algae, microorganisms, animals and biowaste including dead organisms. Second-generation biofuels are produced from non-edible crops and their waste. Second-generation biofuel production has problems such as land insufficiency, soil degradation, loss of vegetation and climate change. Third-generation biofuels are produced from biomass of algae or diatoms through sophisticated technology [13]. Fourth-generation biofuels are an extension work of the third generation, i.e., the production of genetically engineered algae or microorganisms for more biomass and, subsequently, to augment biofuel production. At present, the production of advanced fourth-generation fuels is in progress using non-arable land including electro fuels and photobiological solar fuels [38]. Advanced biofuels including ethanol have advantages over first- and second-generation biofuels because the former causes an increase in food prices and the latter leads to a reduction in greenery in the environment, which contributes to climate change. On the other hand, advanced biofuels could capture solar energy 10 times more efficiently and only a small area is needed to produce sufficient fuel [14]. In advanced biofuel generation technology, algae and microorganisms are grown in special bioreactors supplied with nutrients, organic waste and CO₂ from the air or conventional gas plants [39]. As they grow well in the bioreactors, it is easy to collect fuel from the reactor. Moreover, the advanced method is not laborious when compared with first- and second-generation fuel production processes.

Table 1. Classification of biofuels.

S.No	Class of Biofuel	Involved Organisms
1	First generation	Edible crops: Corn, Sugarcane, Maize, Barley, Oat, etc.
2	Second generation	Non-edible crops: Jatropha, Cotton, Switch grass etc.
3	Third generation	Sea weeds, Algae, Diatoms etc.
4	Fourth generation	Genetically engineered algae, microorganisms etc.

In general, bioethanol is produced from energy crops, agricultural residue, forest residue, sea waste including algae, sewage and municipal waste, which include sugars, starch, cellulose and hemicellulose along with lignocellulosic biomass and other carbohydrates [40]. Specifically, energy crops (switch grass, giant miscanthus etc.), crop residue (corn, sugarcane, maize, barley, oat, sorghum, wheat straw, bagasse, rice straw, coconut shells etc.), plant residue (forest plant residue) and sea sources (seaweeds, algae and diatoms) are widely utilized for ethanol production through fermentation [24,41]. Particularly, a higher amount of ethanol production was achieved with corn in USA and rest of the world [42,43]. In addition, the generation of bioethanol from sugarcane is a regular practice in Brazil and in other parts of the world [44,45]. To date, USA and Brazil are the leading countries in generating ethanol [46]. Figure 2 illustrates the ethanol production levels in various regions of the world (https://en.wikipedia.org/wiki/Ethanol_fuel_by_country, accessed on 10 June 2023). Other sources such as barley, sorghum, maize, rice straw, wheat straw etc. are also used for bioethanol generation [47–50]. The use of agricultural residues limits the reduction of deforestation by decreasing our reliance on forest woody biomass [41]. Agricultural residues, on the other hand, have a short harvest period that renders them more consistently available for ethanol production [51]. Apart from food crops, cotton, jatropha etc. along with forest residues are also used for ethanol production,

which cause climate change [52–54]. Moreover, sewage and municipal waste also contain low amounts of lignocellulosic biomass produced from sawdust, wood chips and bark. Typically, fallen leaves, bark, fruit peel, seed husk etc. are also considered as feedstocks for ethanol production.

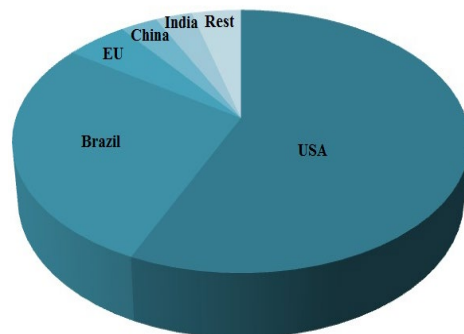


Figure 2. Global ethanol production levels (https://en.wikipedia.org/wiki/Ethanol_fuel_by_country, accessed on 10 June 2023).

Carbohydrates including various sugars, starch and cellulose, including lignocellulose, are the main sources for fermentation. Starch is made up of glucose molecules, and cellulose mainly consists of glucose and hexose. Hemicellulose is generally made up of xylose, arabinose, glucose etc. [55,56]. As compared to grain biomass, cellulosic biomass is a cheap renewable resource and does not compete with the food supply. The degradation of cellulose could be achieved by cellulase and also using other enzymes such as β -glucosidase or exoglucanase, endoglucanases, and cellobiohydrolases or cellobiases [57]. In its non-crystalline form, cellulose is susceptible to enzyme-mediated hydrolysis. For efficient hydrolysis, the accessible area for enzymes should be increased and it is also important to disintegrate the crystalline structure of cellulose [58]. Sugar molecules obtained from cellulose by the enzymes produce lignocellulosic ethanol and reduce the emission of greenhouse gases by 90% when compared with petroleum fuels [59]. Compared with sugar-containing materials, feedstocks having cellulose and starch are cheaper but the cost of cellulase limits the cellulose or starch to fermentable sugar conversion [54]. Lignin, on the other hand, adsorbs enzymes irreversibly and inhibits the action of the enzyme on cellulose chains. Therefore, certain pretreatments are required to reduce the adsorption of enzymes by lignin. Also, the efficient conversion of lignocellulosic materials to ethanol is still a challenging task due to expensive enzymes [60]. However, in order to have an efficient fermentation of lignocellulose-derived substrates, it is necessary to develop strains with a broad substrate-utilization range, capable of using hexose and pentose sugars simultaneously with minimal nutrient requirement, growing at low pH and high temperature [61]. Due to these factors, second-generation feedstock (non-edible crops or forest trees) does not serve the purpose of commercialization of low-cost fuel for transportation and paves the way towards third- and fourth-generation fuel production [62]. Out of all the sources, the best source for ethanol production is algal feedstock due to its easy availability in nature with efficient photosynthetic capacity. In contrast, these algal species are neglected in many places globally. Therefore, the generation of fuels from these feedstocks is also a waste management practice.

4. Effects of Various Factors on Growth and Ethanol Production from Algae

Production of ethanol from various sources, including algae, has certain limitations [63]. Efficient ethanol production from algae completely depends on their growth, accumulation of biomass levels, types of sugars in the feedstock and process of fermentation, which in turn depend on the species and various physicochemical conditions [64]. Specifically, augmentation of biomass in algal culture is one of the key concerns for the improvement of ethanol production. Various physicochemical factors such as light, pH,

temperature, nutrients, vitamins and nanoparticles influence ethanol production in different species. The effects of various factors on algal growth and fermentation were studied elaborately by several researchers [64,65]. Apart from nutrients, light, pH and temperature are the key factors for algal and yeast or bacterial growth (Figure 3).

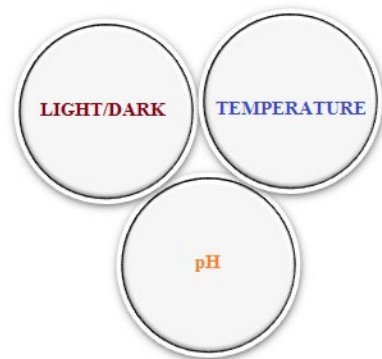


Figure 3. Effect of various physical factors on algal and yeast or bacterial growth.

In general, algae have the capacity for self-fermentation under light and dark conditions. Algae are able to synthesize glucose, starch, cellulose and other carbohydrates in the process of photosynthesis and later catabolize them into various products, including ethanol under anaerobic conditions. The time courses of fermentation under light and dark conditions are compared using different algal feedstocks [23]. The ethanol production levels between light and dark fermentation varied; generally, it was greater under dark conditions. Under dark and anaerobic fermentation, 27% of starch was consumed at 25 °C within 24 h [66]. Thus, fermentation in the dark exhibited better results compared with fermentation under the light, which was proved by several workers [67,68]. Our laboratory found a difference in ethanol production between light and dark fermentations with *Chlorella vulgaris* feedstock using commercial yeast strain [13].

The growth of algae (for feedstock) and yeast or bacteria (for fermentation) depends on the temperature of the culture, which plays a crucial role in biomass and ethanol production. Salmon and Mauricio [69], and Blanch and Clark [70], stated that temperature showed a marked influence on ethanol production based on the strain used. Maximum ethanol production was observed by Bandaru et al. [71] at 32.4 °C from sago starch with *Zymomonas mobilis* MTCC92 strain. Similarly, a high content of ethanol was observed at 30 °C with sugarcane molasses and *Saccharomyces cerevisiae* M30 strain [72]. Ethanol production increased when the fermentation temperature was increased from 25 °C to 30 °C and later gradually decreased with the strain *Saccharomyces cerevisiae* IFST-072011 [73]. Some studies have shown that temperatures above 37 °C are detrimental for ethanol production [65,72]. Our laboratory generated ethanol using the feedstocks of *Chlamydomonas reinhardtii*, *Chlorella vulgaris* and *Chlorococcum minutum* at 27 °C with Baker's yeast [11,13,18,24]. In addition, pH control is an important factor for algal and yeast or bacterial growth, which influences ethanol production. The level of ethanol production completely depends on the type of feedstock (sugar type) as well the fermentation conditions, including the microbial strain [64]. A number of researchers proved that pH 7.0 is best for most of the microalgal growth and development. Bajpai and Margaritis [74] proved that ethanol production was more between 4–6 pH conditions, which indicate the specificity of the fermentation organism. However, ethanol production was strongly suppressed at acidic pH with most of the algal feedstocks. The optimal pH for ethanol production with *Chlamydomonas* feedstock was 7–8 [23]. Yang et al. [75] proved that pH 5.0 resulted in more ethanol yield per dry cell weight. The seeds of *Artocarpus heterophyllus* exhibited flexible ethanol production, and it was noticed that pH 3–5 is better for ethanol fermentation with *Saccharomyces cerevisiae* in the separate fermentation hydrolysis method [76]. From our experience, pH 7.0 is suitable for algal growth and pH 5–6 is best for yeast. Moreover, different types of organisms such

as yeast and bacteria were used to generate ethanol during fermentation, which is also crucial for production levels [24,26].

Nutrients and vitamins play a vital role in the growth and development of algae, including advanced plants. Generally, tris-acetate-phosphate (TAP) medium is used for green algal species, which is universal [77]. Apart from TAP medium, Chu 10 medium, Walne medium, Guillard's F/2 medium etc. are often used for both fresh and marine water algal species. Other than nutrients, most of the photosynthetic organisms, including algae, require vitamins for their growth and development up to certain levels [13,18]. Few studies have been conducted on the generation of ethanol from bacteria and yeast with the help of vitamins [25,78]. Augmentation of ethanol production was achieved with cellulose obtained from media containing yeast extract and vitamin B₁₂ by *Clostridium thermocellum* I-1-B. A double dose of vitamins and the exponential feeding of these vitamins increased the ethanol concentrations [25]. Tandon et al. [79] proved that productivity of algae is enhanced through exogenous supply of vitamins. Recently, Ruangsomboon et al. [80] used B1, B7 and B12 vitamins for the enhancement of growth and biomass of green alga *Botryococcus braunii* in turn biofuels and succeeded. For the first time, enhancement of ethanol production was carried out using vitamin-assisted (B₁, B₇ and B₁₂) algal feedstocks in our laboratory [18]. Also, several nanoparticles are being screened at present to improve the biomass, and thus, ethanol production from algal species. Recently, we improved the biomass in green alga, *Chlorococcum minutum*, using graphene-oxide-supported platinum–ruthenium (Pt-Ru/RGO) nanoparticles along with TAP medium and achieved the augmentation of ethanol production by fermentation [24].

5. Production of Bioethanol from Various Algal Species

Algae are primitive photosynthetic plants that prevail in almost all existing earth ecosystems and have the capability to produce several beneficial products due to their photosynthetic capacity and ability to grow in non-potable water sources [64]. Both marine and freshwater algal species are available abundantly with uni and multicellular natures. Based on size, algae are broadly classified as macro and microalgae [13,14]. Both macro and microalgae are the most promising feedstocks due to their rich biomass, which is easy to convert into ethanol. Algae are further classified into subgroups namely, green (chlorophyceae), red (rhodophyceae), brown (phaeophyceae) and blue–green algae (cyanophyceae) based on their pigment content [81]. Green algae are popular species for bioethanol production due to their potential photosynthetic pathway, which leads to generating more carbohydrates and sugars [65]. Generation of ethanol from algal feedstock has been an interesting research topic in recent days. A number of works on ethanol production using various algal feedstocks have been carried out [7,82]. Figure 4 illustrates the schematic representation of ethanol production from algal species.

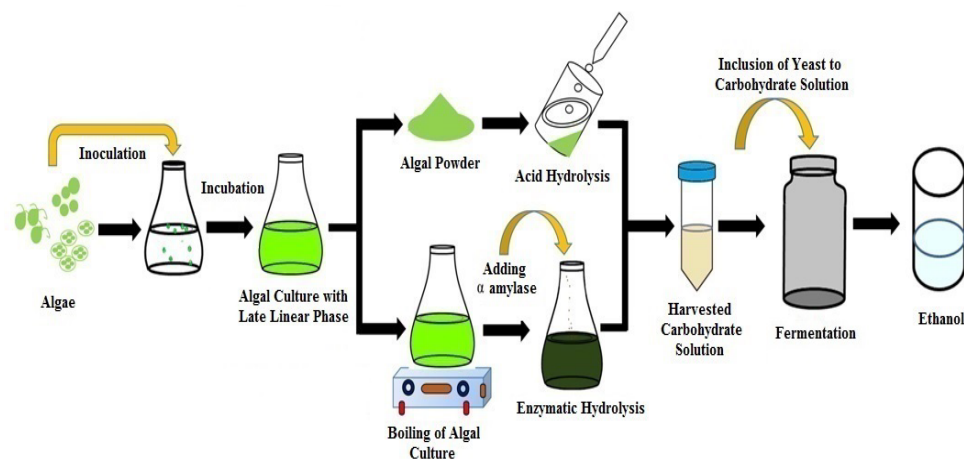


Figure 4. Schematic representation of ethanol production from algae and their waste.

5.1. Production of Ethanol from Green Algae

Most of the biofuel works have been carried out with green algal species including the model alga, *Chlamydomonas reinhardtii*. Table 2 represents the green algal species involved in ethanol production. Ethanol production from carbohydrate-rich microalgae through intracellular anaerobic fermentation was an alternative to the conventional fermentation process. Both processes were practiced using *Chlorella vulgaris* and *Chlamydomonas reinhardtii*, and the earlier one proved to be the best with respect to energy content in conventional fermentation [23]. Ueno et al. [83] generated ethanol using *Chlorococcum littorale*, a marine green alga, by the process of dark fermentation. Overall, the maximum output was 450 $\mu\text{mol/g}$ dry weight at 30 °C, and temperature influences the starch degradation. The relation between carbon dioxide fixation and ethanol production by dark fermentation was established by screening of 200 algal strains [84]. A hyperactive strain, namely, *Chlamydomonas* sp. YA-SH-1, was found to be excellent with respect to growth and ethanol production by converting the starch. Ethanol fermentation work was carried out using green microalgae grown in a solution from the desalting process of soy sauce waste treatment by Shirai et al. [85]. A total 11 mg of ethanol was produced from 1 g of *Dunaliella* cells.

Table 2. Production of ethanol from different green algal species.

S.No	Ethanol Production Work	Name of the Algae	Reference
1	Both conventional and intracellular anaerobic fermentation were practiced using microalgae, and the former generated more ethanol	<i>Chlorella vulgaris</i> , <i>Chlamydomonas reinhardtii</i>	Hirano et al., 1997 [23]
2	Using the marine alga, ethanol was generated under dark fermentation	<i>Chlorococcum littorale</i>	Ueno et al., 1998 [83]
3	Screening for carbon dioxide fixation; subsequently, ethanol production by fermentative microalgae was carried out	200 strains including <i>Chlamydomonas</i> YA-SH-1	Hirayama et al., 1998 [84]
4	Ethanol fermentation studies using microalgae grown in a solution from the desalting process of soy sauce waste treatment	<i>Chlamydomonas</i> sps <i>Dunaliella</i> sps, <i>Nannochloropsis oculata</i> , <i>Tetraselmis tetrathele</i>	Shirai et al., 1998 [85]
5	The saccharification of marine microalgae was carried out for ethanol production using the bacterium <i>Pseudoalteromonas undina</i> NKMB 0074, which possesses the marine amylase	Green microalgae (NKG120701)	Matsumoto et al., 2003 [86]
6	The production of ethanol using the algae was established	Different algae (Zygnemataceae, Cladophoraceae, Oedogoniales etc.)	Bush and Hall, 2006 [87]
7	Algal biomass was pretreated with sulfuric acid for glucose release and, in turn, ethanol production	<i>Chlamydomonas reinhardtii</i> UTEX 90	Nguyen et al., 2009 [88]
8	Ethanol was produced using the biomass of green alga with enzymatic pretreatment	<i>Chlamydomonas reinhardtii</i> UTEX 90	Choi et al., 2010 [89]
9	Ethanol production carried out using the microalgal biomass	<i>Chlorococcum</i>	Harun et al., 2010 [90]
10	NaOH treatment was carried out for algal biomass for bioethanol production	<i>Chlorococcum infusionum</i>	Harun et al., 2011 [91]
11	Green alga was used for ethanol production through fermentation by both bacterium and yeast (<i>Zymomonas mobilis</i> and <i>Saccharomyces cerevisiae</i>)	<i>Spirogyra</i>	Sulfahri et al., 2011 [26]
12	Algal species was used for ethanol production; <i>Aspergillus niger</i> MTCC 2196 and <i>Saccharomyces cerevisiae</i> MTCC 170 were used for saccharification and fermentation	<i>Spirogyra</i>	Eshaq et al., 2011 [92]
13	Ethanol was extracted from marine algae carbohydrates using <i>E. coli</i> strains	<i>Undaria pinnatifida</i> , <i>Chlorella vulgaris</i> , <i>Chlamydomonas reinhardtii</i>	Lee et al., 2011 [93]

Table 2. Cont.

S.No	Ethanol Production Work	Name of the Algae	Reference
14	Improvement of ethanol production was noticed with sonicated green alga cultivated in swine wastewater effluent.	<i>Scenedesmus obliquus</i> YSW15	Choi et al., 2011 [94]
15	Acid-pretreated microalgal biomass enhanced the bioethanol production	<i>Chlorococcum humicola</i>	Harun and Danquah, 2011 [95]
16	Improved ethanol production was observed by enzymatic hydrolysis of microalgal biomass	<i>Chlorococcum</i> sps	Harun and Danquah, 2011 [96]
17	Acid-treated dried biomass of <i>Scenedesmus obliquus</i> resulted efficient sugar extraction	<i>Scenedesmus obliquus</i>	Mirinda et al., 2012 [97]
18	Ethanol production was achieved by the conversion of hydrolysates of algal species with <i>Zymomonas mobilis</i>	<i>Spirogyra hyalina</i>	Sulfhari et al., 2012 [98]
19	Ethanol production achieved through the influence of photobioreactors and culture conditions with <i>Scenedesmus obliquus</i>	<i>Scenedesmus obliquus</i>	Miranda et al., 2012 [99]
20	Optimal starch extraction by ultrasonic homogenizer and ethanol production through conversion withSSF method	<i>Chlamydomonas fasciata</i> Ettl 437	Asada et al., 2012 [100]
21	Ethanol production was carried out using different hydrolysis and fermentation methods	<i>Chlorella vulgaris</i>	Ho et al., 2013 [101]
22	Bioethanol was generated from a green alga by enzymatic saccharification along with yeast	<i>Dunaliella tertiolecta</i>	Lee et al., 2013 [102]
23	Ethanol production was carried out using the cake of green alga along with cost of the experiment and environmental safety aspects	<i>Chlorella vulgaris</i>	Moncada et al., 2013 [103]
24	Sugars were released due to virus infection in green alga, which were used for bioethanol production through <i>E.coli</i>	<i>Chlorella variabilis</i> NC64A	Cheng et al., 2013 [104]
25	Eighteen algal strains isolated in Pearl River delta were used for ethanol production; pretreatment was carried with dilute acid and cellulase	<i>Mychonastes afer</i> PKUAC 9 and <i>Scenedesmus abundans</i> PKUAC 1	Guo et al., 2013 [39]
26	Five pretreatments were conducted with macroalgal biomass for ethanol production	<i>Chaetomorpha linum</i>	Schultz-Jensen et al., 2013 [105]
27	Microalgae cultivated in two stages for more carbohydrates, which were used for ethanol production	<i>Scenedesmus obliquus</i> CNW-N	Ho et al., 2013 [106]
28	Macrophytic green alga was used for ethanol production through enzymatic hydrolysis	<i>Ulva fasciata</i>	Trivedi et al., 2013 [107]
29	Green alga from Hawaii was used for ethanol production by saccharification cellulase of <i>Trichoderma reesei</i> followed by fermentation	<i>Ulva reticulata</i>	Yoza and Masutani, 2013 [108]
30	Wall-deficient mutant model alga used for ethanol production by acid hydrolysis and fermentation of starch through yeast	<i>Chlamydomonas reinhardtii</i> mutant <i>cw15</i>	Scholz et al., 2013 [109]
31	Less ethanol was generated in the cultures without nitrogen content	<i>Chlorella vulgaris</i>	Salman and Ali, 2014 [27]
32	Improved ethanol was noticed from the nutrient stress-induced microalga by enzymatic hydrolysis and immobilized yeast fermentation	<i>Chlorella vulgaris</i>	Kim et al., 2014 [68]
33	Ethanol production was carried out with filamentous green alga	<i>Tribonema</i> sps	Wang et al., 2014 [110]
34	Green alga biomass particle size was altered and small particles resulted more ethanol	<i>Chlorococcum infusionum</i>	Harun et al., 2014 [111]
35	Physical, chemical and enzymatic pretreatments for saccharification of carbohydrates were practiced for ethanol production in microalgal species	<i>Chlorella sorokiniana</i> , <i>Nannochloropsis gaditana</i>	Hernandez et al., 2015 [112]

Table 2. Cont.

S.No	Ethanol Production Work	Name of the Algae	Reference
36	Ethanol was generated from green alga after lipid extraction	<i>Chlorella</i> sps KR-1	Lee et al., 2015 [113]
37	Ethanol production was achieved in a marine alga using two fermentation organisms, i.e., <i>Pachysolen tannophilus</i> and <i>Zymomonas mobilis</i> , and the earlier one resulted more yield	<i>Ulva rigida</i>	El Harchi et al., 2015 [114]
38	Both ethanol and diesel were prepared using a green microalga cultivated in tubular photobioreactor	<i>Scenedesmus bijugatus</i>	Ashokkumar et al., 2015 [115]
39	Ethanol generation was studied using model alga and also using a hybrid system	<i>Chlamydomonas reinhardtii</i>	Costa et al., 2015 [116]
40	Ethanol production and heavy metal removal was carried out using green alga grown in various concentrations of wastewater	<i>Scenedesmus obliquus</i>	Hamouda et al., 2016 [117]
41	Ethanol was produced from green alga using <i>Zymomonas mobilis</i>	<i>Spirogyra hyalina</i>	Sulfhari et al., 2016 [118]
42	Ethanol and methane were produced from a green macroalga using a biorefinery concept	<i>Chaetomorpha linum</i>	Yahmed et al., 2016 [119]
43	Green alga was tested for ethanol production using HCl hydrolysis and yeast fermentation	<i>Chlorella vulgaris</i>	Bhooshan Kumar et al., 2016 [120]
44	Ethanol was produced using various experimental conditions and pretreatments from halophilic microalgal biomass	<i>Dunaliella</i> sps	Karatay et al., 2016 [121]
45	Efficient ethanol was produced from marine alga in a pilot scale model	<i>Dunaliella tertiolecta</i>	Varela-Bojórquez et al., 2016 [122]
46	Both <i>Cyclotella</i> and green algal species generated more ethanol in SHE treatment in yeast fermentation	<i>Chlorella vulgaris</i> YSL001 <i>Uronema belkai</i>	Hwang et al., 2016 [123]
47	Repeated batch SSF was proved as best method for long-term ethanol production with microalgal biomass using immobilized <i>Saccharomyces cerevisiae</i>	<i>Chlamydomonas mexicana</i>	El-Dalatony et al., 2016 [124]
48	Improvement of ethanol generation was observed with green alga through chemical and biological saccharification	<i>Ulva fasciata</i>	Hamouda et al., 2016 [125]
49	Enhancement of enzymatic saccharification of algal biomass resulted more ethanol production	<i>Chaetomorpha linum</i>	Neifar et al., 2016 [126]
50	Hungate technique for ethanol fermentation of algae using <i>Saccharomyces cerevisiae</i>	<i>Spirogyra hyalina</i>	Sulfahri et al., 2016 [127]
51	Algal strains grown under nitrogen deficiency act as potential feedstock for ethanol production	17 strains <i>Desmodesmus</i> sps SP2-3	Rizza et al., 2017 [128]
52	Efficient ethanol production was noticed from green alga grown in plantain peel extract with increased fermentation time	<i>Chlorella vulgaris</i>	Agwa et al., 2017 [129]
53	Pretreated green microalga generated more and cost-effective ethanol	<i>Scenedesmus dimorphus</i>	Chng et al., 2017 [130]
54	Cultivation of marine microalgae in treated municipal wastewater toward bioethanol production	<i>Nannochloropsis oculata</i> , <i>Tetraselmis suecica</i>	Reyimu and Ozcimen, 2017 [131]
55	Ethanol was produced from defatted residues of freshwater green alga along with biodiesel	<i>Cosmarium</i> sps	Manoj et al., 2018 [12]
56	Green alga was used for concomitant production of both ethanol and diesel	<i>Scenedesmus</i> sp.	Ramachandran and Incharoensakdi, 2018 [132]
57	Both micro- and macroalgal species were used for ethanol production through different hydrolysis and fermentation methods	<i>Ulva fasciata</i> , <i>Chlorella vulgaris</i>	Hamouda et al., 2018 [8]
58	Enhanced ethanol was obtained through acid hydrolysis of green alga	<i>Ulva rigida</i>	El Harchi et al., 2018 [133]

Table 2. Cont.

S.No	Ethanol Production Work	Name of the Algae	Reference
59	Bioethanol was generated from green alga by acidpre-treatment on the microbial fermentation process	<i>Chlorella</i> sps	Phwan et al., 2019 [134]
60	Effect of hydrolysis time and acid concentration on ethanol generation from microalga	<i>Scenedesmus</i> sps	Agustini et al., 2019 [135]
61	Augmented ethanol production from green algal biomass by hydrothermal pretreatment and enzymatic hydrolysis	<i>Chlorella</i> sps	Ngamsirisomsakul et al., 2019 [136]
62	Ethanol produced through fed-batch method using green algal species	<i>Scenedesmus raciborskii</i>	Alam et al., 2019 [137]
63	Ethanol production from algal species using <i>Saccharomyces cerevisiae</i> , <i>Pichia kudriavzevii</i> and <i>Kluyveromyces thermotolerans</i>	<i>Spirogyra peipingensis</i>	Sulfahri et al., 2019 [138]
64	Ethanol production from acid-pretreated microalgal hydrolysate using microwave-assisted heating wet torrefaction	<i>Chlorella vulgaris</i> ESP-31	Yu et al., 2020 [139]
65	Ethanol production from green seaweed residue	<i>Codium tomentosum</i>	Gengaiah et al., 2020 [140]
66	Ethanol from green algal feedstock using both light and dark fermentation	<i>Chlorella vulgaris</i>	Varaprasad et al., 2020 [13]
67	Different hydrolysis and post-treatment processes of microalgal hydrolysate on bioethanol production	<i>Chlorella</i> ABC-001	Seon et al., 2020 [141]
68	Supplementation of lysine and magnesium resulted more carbohydrates and, in turn, ethanol content	<i>Scenedesmus acuminatus</i>	Chandra et al., 2020 [142]
69	Production of ethanol in green alga using efficient saccharification of biomass by <i>Trichoderma harzianum</i> enzymes	<i>Chlamydomonas reinhardtii</i>	Bader et al., 2020 [143]
70	Bioethanol was produced using deoiled biomass of green alga using eco-friendly bio-based heterogeneous catalysts and hydrolysates	<i>Scenedesmus obliquus</i>	Gohain et al., 2021 [144]
71	Influence of different factors on biomass, bioethanol and biohydrogen generation in green alga	<i>Chlamydomonas reinhardtii</i>	Ragasudha et al., 2021 [11]
72	Vitamins improved the ethanol in green alga	<i>Chlorococcum minutum</i>	Varaprasad et al., 2021 [18]
73	Limiting the nitrate content in algal culture enhanced the total carbohydrate and starch, and thus, ethanol production	<i>Chlorella sorokiniana</i>	Kaur et al., 2022 [145]
74	Ethanol generation from green alga through Pt-Ru/RGO nanoparticles	<i>Chlorococcum minutum</i>	Varaprasad et al., 2022 [24]
75	Microalgae biomass was used for ethanol generation at high-solids loadings	<i>Chlorella vulgaris</i> FSP-E	Condor et al., 2022 [146]
76	Ethanol production from green alga grown in unsterilized swine wastewater	<i>Chlorella vulgaris</i> ESP-31 and <i>Chlorella sorokiniana</i> AK-1	Acebu et al., 2022 [147]
77	Process and technoeconomic analysis of ethanol production from residual biomass of marine macroalga	<i>Ulva lactuca</i>	Gengaiah et al., 2023 [148]
78	Production and use of biomass in synthetic municipal wastewater for integrated biorefineries	<i>Scenedesmus acuminatus</i> CCALA436	Onay and Aladag, 2023 [7]

Matsumoto et al. [86] conducted the saccharification of marine green microalgae (NKG 120701) using marine bacteria, namely, *Pseudoalteromonas undina* NKMB 0074. In this process, the amylase of marine bacteria is useful for carbohydrate breakdown in saline conditions. Establishment of an ethanol production protocol using yeast under dark and anaerobic conditions was performed with colony-forming green algal species belonging to the families zygnemataceae, cladophoraceae and oedogoniaceae [87]. The remaining biomass will be useful for biodiesel production as a starting material. The model alga, *Chlamydomonas reinhardtii*, biomass was pretreated with acid for glucose release, which is useful for improving ethanol production [88]. The treatment was carried out with 1–5% of sulfuric acid at 100 to 120 °C for 15 to 120 min. Later, the pretreated slurry was used for ethanol production by utilizing yeast (*Saccharomyces cerevisiae* S288C). Choi et al. [89]

generated ethanol using the feedstock of *Chlamydomonas reinhardtii* UTEX 90 strain with two commercial hydrolytic enzymes; one gram of biomass generated 235 mg of ethanol in the process of the separate hydrolysis and fermentation method. A microalga, *Chlorococum*, was used as the feedstock for ethanol production under different fermentation conditions [90]. *Saccharomyces bayanus* was used as a yeast source for fermentation (Table 2).

Alkaline pretreatment using NaOH was carried out with *Chlorococcum infusionum* for ethanol production [91]. At 0.75% of NaOH, 0.26 g of ethanol per gram biomass was generated at 120 °C for 30 min. Sulphari et al. [26] used *Spirogyra* feedstock for bioethanol production with both *Zymomonas mobilis* (bacterium) and *Saccharomyces cerevisiae* (yeast). More ethanol production was noticed with *Zymomonas mobilis* than yeast, with a faster rate. In another work, similar *Spirogyra* dry powder was used for saccharification and simultaneous fermentation by *Aspergillus niger* MTCC 2196 and *Saccharomyces cerevisiae* MTCC 170 for ethanol production [92]. Ethanol production was carried out with marine algae using different strains of *Escherichia coli* [93]. The highest amount of ethanol (0.4 g/g biomass) was generated with pretreated *C. vulgaris* and *E. coli* SJJ2526. The sonicated samples of *Scenedesmus obliquus* YSW15 cultivated in swine wastewater effluent released more biomass, which improved the ethanol production [94]. Harun et al. [95] noticed that the acid-pretreated *Chlorococcum humicola* biomass released more sugars, which augmented the ethanol production. The same group worked on enzymatic hydrolysis of *Chlorococcum* using the cellulase obtained from *Trichoderma reesei* ATCC 26921. The hydrolysis improved the saccharification process of algal biomass, which enhanced the ethanol production [96]. An improvement in ethanol production was achieved with acid hydrolysis of dried powder of *Scenedesmus obliquus* [97]. In this method, they used 2N H₂SO₄ at 121 °C for 30 min, which enhanced the sugar extraction with low cost.

Enhancement of ethanol production was achieved with hydrolysates of *Spirogyra hyalina* using different pH, gases and fermentation duration in [98]. Fermentation was carried out with the bacterium, *Zymomonas mobilis*. Effects of photobioreactors (PBRs) and light conditions on ethanol production from *Scenedesmus obliquus* were studied by Miranda et al. [99]. Efficient starch extraction was achieved from *Chlamydomonas fasciata* Ettl 437 with the influence of 30 min ultrasonic treatment by a homogenizer [100] followed by simultaneous saccharification and fermentation. Overall, 0.194 and 0.168 g-ethanol/g-dry microalgae were obtained with and without yeast (*Saccharomyces cerevisiae* AM12), respectively. Augmented ethanol production was noticed with the feedstock of *Chlorella vulgaris* FSP-E using various hydrolysis and fermentation methods, which include both SSF and SHF [101]. The biomass of the green alga *Dunaliella tertiolecta* was used to generate ethanol through enzymatic saccharification and later by fermentation using *Saccharomyces cerevisiae* [102]. Moncada et al. [103] used the cake of *Chlorella vulgaris* for ethanol production through enzymatic hydrolysis and fermentation by utilizing *Saccharomyces cerevisiae*; also, they worked on economic assessment and safety aspects of ethanol production from algal species. Virus infection in *Chlorella variabilis* NC64A leads to the disruption and hydrolysis of algal biomass, which consequently released the sugars [104]. Ethanol was generated with these sugars through fermentation by *Escherichia coli* KO11. Guo et al. [39] identified 18 algal strains from Pearl River delta, which are used for ethanol production. Efficient ethanol production was achieved with *Scenedesmus abundans* PKUAC12 through pretreatment with dilute acid and cellulase. Schultz-Jensen et al. [105] applied hydrothermal pretreatment, steam explosion, ball milling, wet oxidation and plasma-assisted pretreatment to the biomass of *Chaetomorpha linum* for ethanol production. Both wet oxidation and ball milling resulted more ethanol production with this green macroalga (Table 2).

Two-stage cultivation conditions were applied to *Scenedesmus obliquus* CNW-N cultures for carbon dioxide fixation for more carbohydrates [106]. Further, these feedstocks were used for ethanol production through the separate hydrolysis and fermentation method. Trivedi et al. [107] generated ethanol using *Ulva fasciata* (green macrophytic alga) with the help of enzymatic hydrolysis. Specifically, cellulase 22119 converted the biomass into reducing sugars. Yoza and Masutani [108] used *Ulva reticulata*, a green alga isolated from the

region of Hawaii, for ethanol production. In this process, they followed the enzymatic saccharification using the cellulase of *Trichoderma reesei* followed by fermentation. The biomass of wall-deficient cells of mutant *Chlamydomonas reinhardtii* was treated with 70 °C ethanol and 60 °C hexane before sulfuric acid hydrolysis [109]. The hydrolyzed starch generated ethanol in the process of fermentation by using *Saccharomyces cerevisiae*. Ethanol production was carried out with *Chlorella vulgaris* grown in various nitrogen concentrations [27]. The highest ethanol content was observed with 0.0 g/L nitrogen, and more protein content was noticed with a 12.0 g/L dose. Similarly, nitrogen limitation induced more carbohydrate production in *Chlorella vulgaris* [68]. Further, these carbohydrates were used for ethanol production through enzymatic hydrolysis and immobilized yeast fermentation. The filamentous green alga *Tribonema* species was used for bioethanol production [110]. Sulfuric acid was used for hydrolysis of wall cells into sugars; later, fermentation was carried out by *Saccharomyces cerevisiae*. Further, the same algal biomass was used for biodiesel production. Harun et al. [111] altered the particulate size of biomass of *Chlorococcum infusionum*, which affects the hydrolysate properties. Small-sized particles resulted more bioethanol production (Table 2).

The biomass of both *Chlorella sorokiniana* and *Nannochloropsis gaditana* were used for ethanol production through saccharification of carbohydrates by physical, chemical and enzymatic pretreatments [112]. Carbohydrate-enriched *Chlorella* species KR-1 was used for ethanol production after lipid extraction [113]. El Harchi et al. [114] generated ethanol using the biomass of marine *Ulva rigida* with two fermentation organisms, i.e., *Pachysolen tannophilus* and *Zymomonas mobilis*. The highest ethanol content was noticed with *Pachysolen tannophilus* fermentation. Ashokkumar et al. [115] worked on both ethanol and diesel production from the green microalga *Scenedesmus bijugatus*. Alga was cultivated in a tubular photobioreactor and lipid-extracted biomass was used for ethanol generation. Costa et al. [116] investigated the initial inoculum and carbon source for *Chlamydomonas reinhardtii* in tris-acetate phosphate medium without sulfur as well as ethanol production. They followed a hybrid system by using both *Chlamydomonas reinhardtii* and *Rhodobacter capsulatus* for ethanol generation. *Scenedesmus obliquus* grown in various concentrations of wastewater was used for ethanol production by fermentation [117]. This species is also useful for the removal of heavy metal, which is one of the remediation processes. Sulphari et al. [118] used the biomass of *Spirogyra hyalina* for ethanol production by altering the heating duration, pH and fermentation duration; they used *Zymomonas mobilis* for the fermentation process. A green alga, *Chaetomorpha linum*, was used for the production of ethanol through enzymatic hydrolysis and fermentation by *Saccharomyces cerevisiae* [119]; the study also used the biomass for methane production (Table 2).

Bhooshan Kumar et al. [120] worked on the biomass of *Chlorella vulgaris* to determine its efficiency in ethanol production. Hydrolysis was carried out with HCl and fermentation was performed using *Saccharomyces cerevisiae*. A high content of ethanol was obtained of around 13.2 wt% at 30 °C for 28 h. The feedstock of *Dunaliella* was used for the production of ethanol by applying various pretreatment methods in [121]. They followed certain methods such as alteration in nutrients, dilute acid hydrolysis, fermentation pH ranges and incubation duration to obtain maximum ethanol yield. Similarly, the biomass of *Dunaliella tertiolecta* was used for ethanol production in a pilot scale model, where the species was found to be ideal [122]. The feedstocks of green algae such as *Chlorella vulgaris* YSL001 and *Uronema belkai* along with *Cyclotella* (diatom) were used for ethanol production [123]. High ethanol content was obtained with combined sonication, heat and enzyme (SHE) treatment along with yeast (*Dekkera bruxellensis*) fermentation. Repeated-batch simultaneous saccharification and fermentation with immobilized *Saccharomyces cerevisiae* was proved as an efficient ethanol production method with *Chlamydomonas mexicana*; before that, the algal biomass was treated with both sonication and acid hydrolysis [124]. Hamouda et al. [125] achieved augmented ethanol generation with *Ulva fasciata* through chemical and biological saccharification. Improved enzymatic hydrolysis of the biomass of *Chaetomorpha linum* with the cellulase of *Aspergillus niger* resulted in augmented ethanol production by

Saccharomyces cerevisiae in fermentation [126]. The evaluation of sugar levels was carried out with a three-factor, three-level Box–Behnken design. The Hungate technique was applied during fermentation for *Spirogyra hyalina* cultures to improve the ethanol production [127]. They used two gases, i.e., nitrogen and hydrogen; three pH ranges, i.e., 4, 5 and 6; and four fermentation intervals, i.e., 0, 24, 48 and 72 h. Ethanol content was greater with hydrogen gas at pH 4 after 72 h.

Microalgae grown under nitrogen deficiency in laboratory conditions acts as a potential feedstock for fermentation and a hub for sourcing bioethanol [128]. *Chlorella vulgaris* was cultivated with plantain peel extract for 14 days as carbon source [129]. Later, the biomass was treated with both acid and enzyme hydrolysis. Further, these hydrolysates were fermented by *Saccharomyces cerevisiae* and *Aspergillus* species using SHF and SHCF methods. Higher ethanol content was noticed with increased fermentation time. A carbohydrate-rich green microalga, *Scenedesmus dimorphus*, was cultivated under carbon-dioxide-enriched air. In addition, the biomass was pretreated with various methods and fermentation was carried out to obtain cost-effective bioethanol [130]. Both *Nannochloropsis oculata* and *Tetraselmis suecica* were cultivated on treated municipal wastewater [131]. Municipal-wastewater-grown *Tetraselmis suecica* was found to be better for ethanol production than the former one. Manoj et al. [12] worked on ethanol production from defatted residues of *Cosmarium* (freshwater green alga), using *Saccharomyces cerevisiae* during fermentation. Further, they used this alga for biodiesel production. Similarly, both bioethanol and biodiesel were produced from the feedstock of *Scenedesmus* species using a biorefinery approach, i.e., direct transesterification was performed followed by fermentation [132]. Both macro- and microalgal species, i.e., *Ulva fasciata* and *Chlorella vulgaris*, were used for ethanol production through different hydrolysis and fermentation processes [8]. Thermal acid hydrolysis was optimized for improvement of sugars in *Ulva rigida*. Further, these sugars were converted into ethanol by *Pachysolen tannophilus* during fermentation [133]. Augmented production of ethanol was achieved from *Chlorella* species through acid pre-treatment and microbial fermentation [134]. The effect of hydrolysis time and acid concentration was investigated in microalgae by Augustini et al. [135] for the improvement of ethanol production (Table 2).

Enhancement of ethanol generation from the biomass of *Chlorella* species was achieved by hydrothermal pretreatment and enzymatic hydrolysis in [136]. Alam et al. [137] used the biomass of *Scenedesmus raciborskii* WZKMT for ethanol production by following fed-batch enzymatic hydrolysis and fermentation processes. A maximum of 79.38 g ethanol per liter was achieved with the repeated fed-batch process. *Spirogyra peipingensis* was found to be an efficient raw material for bioethanol production [138]. The authors used *Saccharomyces cerevisiae*, *Pichia kudriavzevii* and *Kluyveromyces thermotolerans* for fermentation, and the first one generated more ethanol. Acid-pretreated *Chlorella* hydrolysate generated more ethanol via microwave-assisted heating wet torrefaction [139]. The biomass of *Codium tomentosum* (green seaweed) was utilized as a renewable feedstock for ethanol production [140]. Bioethanol production from the feedstock of *Chlorella vulgaris* was achieved via light and dark fermentation in our laboratory [13]. A high amount of ethanol production was noticed in dark fermentation. Seon et al. [141] produced ethanol using different hydrolysis and post-treatment processes with *Chlorella* species ABC-001. Hydrolysis using $\text{H}_2\text{SO}_4 + \text{Ca}(\text{OH})_2$ and fermentation through *Saccharomyces cerevisiae* KL17 yielded augmented ethanol generation. Supplementation of lysine and magnesium resulted in more carbohydrate and, thus, more ethanol content in *Scenedesmus acuminatus* [142]. Specifically, the inclusion of lysine at alkaline pH improved the biomass and carbohydrate contents (Table 2).

Enhancement of ethanol production was achieved with the biomass of *Chlamydomonas reinhardtii* using an efficient saccharification method [143]. The authors used *Trichoderma harzianum* enzymes, which efficiently convert complex sugars into simple sugars. Gohain et al. [144] worked on deoiled *Scenedesmus obliquus* waste for ethanol production using various eco-friendly heterogeneous catalysts. The obtained hydrolysate yielded a high amount of ethanol (68.32% at 8.24 g/L) during fermentation by *Saccharomyces cerevisiae*. Our laboratory improved the ethanol production levels in *Chlamydomonas reinhardtii* by adding

sodium bicarbonate in TAP medium [11]. Also, our laboratory augmented the ethanol content through the exogenous supply of vitamins in the green alga *Chlorococcum minutum* [18]. A combination of B1, B7 and B12 vitamins in the algal cultures improved the biomass and, thus, the ethanol production. Enhancement of ethanol production in *Chlorella sorokiniana* was achieved by limiting the nitrate content in the culture medium [145]. Recently, our laboratory used Pt-Ru/RGO nanoparticles in TAP medium, which improved the biomass and ethanol production in *Chlorococcum minutum* [24]. Probably, these nanoparticles are involved in the electron transport chain in photosynthesis, which enhances the carbon dioxide fixation. Condor et al. [146] produced a high amount of ethanol through the fermentation of *Chlorella vulgaris* FSP-E hydrolysate under high-solids loadings with *Saccharomyces cerevisiae* FAY-1. *Chlorella vulgaris* ESP-31 grown in unsterilized swine wastewater resulted in more ethanol production. The swine wastewater is a cost-effective source of nutrients for the algal species. Initially, both *Chlorella sorokiniana* AK-1 and *Chlorella vulgaris* ESP-31 were grown in this wastewater [147]. Gengaiah et al. [148] selected the biomass of *Ulva lactuca* for ethanol production through the separate hydrolysis and fermentation method. Further, generated ethanol was characterized by GC-MS. Very recently, Onay and Aladag [7] used *Scenedesmus acuminatus* CCALA436 grown in wastewater and mepiquat chloride for the production of ethanol (Table 2). When compared with green algal species, the other groups of algal species are less focused with respect to ethanol production due to their variation in carbohydrate content.

5.2. Generation of Ethanol from Brown Algae

Brown algal species are a source for various bioproducts, including fuels. Among all brown algal species, *Laminaria* or *Saccharina* and *Sargassum* are widely used for ethanol production. Ethanol production was carried out using feedstock of a brown alga, *Saccharina latissima*, with the assistance of laminarinase and *Saccharomyces cerevisiae* [149] (Table 3). The hydrolysate of *Sargassum sagamianum* was used for ethanol production with six yeast (*Pichia stipitis*) strains [150]. Out of six strains, ATCC 7126, ATCC 58784 and ATCC 58376 generated three to eight times more ethanol. The bioethanol generation processes for brown alga (*Saccharina japonica*) were simulated using the Aspen Plus V7.3 software [151]. This was the first simulation work on algal ethanol production, which will be useful for commercial practice. Lee and Lee [152] tested eight different yeast strains for ethanol production from the feedstock of *Laminaria japonica*. Among all strains, the *Saccharomyces cerevisiae* strain KCCM50550 generated more ethanol, i.e., 2.59 g/L from 10.0 g/L of mannitol. The biomass of *Saccharina japonica* was used for ethanol production with low-concentration acid pretreatment, i.e., 0.06% sulfuric acid at 170 °C for 15 min [153]. Further, simultaneous saccharification and fermentation were carried out with cellulase and β -glucosidase along with *Saccharomyces cerevisiae* DK 410362. Borines et al. [154] used the biomass of the *Sargassum* species for ethanol production through acid pretreatment and fermentation by *Saccharomyces cerevisiae*. Ethanol production was achieved using the biomass of brown macroalgae through the re-engineering of alginate and mannitol catabolic pathways along with fermentation by a synthetic yeast platform [155].

Table 3. Production of ethanol from various brown algal species.

S.No	Ethanol Production Work	Name of the Algae	Reference
1	Fermentation of brown alga was carried out for ethanol production considering various pretreatments	<i>Saccharine lasissima</i> (<i>Laminaria</i>)	Adams et al., 2009 [149]
2	A brown alga hydrolysate was used to generate bioethanol with six <i>Pichia stipitis</i> strains	<i>Sargassum sagamianum</i>	Ji-Hyeon et al., 2010 [150]
3	Simulations of processes of bioethanol generation were conducted using brown alga	<i>Saccharina japonica</i>	Fasahati and Liu, 2012 [151]
4	Various yeast strains were used for ethanol production from brown algae	<i>Laminaria japonica</i>	Lee and Lee, 2012 [152]

Table 3. Cont.

S.No	Ethanol Production Work	Name of the Algae	Reference
5	Brown alga was used to generate ethanol through low-acid pretreatment followed by simultaneous saccharification and fermentation	<i>Saccharina japonica</i>	Lee et al., 2013 [153]
6	Macroalgal species was used for ethanol production through acid pretreatment and fermentation	<i>Sargassum</i> sps	Borines et al., 2014 [154]
7	Improved ethanol production from brown macroalgae sugars was achieved by a synthetic yeast platform along with re-engineering of catabolic pathways	Brown macroalgae	Enquist-Newman et al., 2014 [155]
8	Both ethanol and protein generation were carried out from brown alga	<i>Laminaria digitata</i>	Hou et al., 2015 [156]
9	Ethanol was produced by mixing the saccharified waste of brown alga and paper shredder scrap using 19 strains of <i>Saccharomyces cerevisiae</i>	<i>Undaria pinnatifida</i>	Obara et al., 2015 [157]
10	Brown seaweed was used for ethanol using non-conventional yeasts	<i>Ascophyllum nodosum</i> and <i>Laminaria digitata</i>	Obata et al., 2016 [158]
11	Brown marine macroalgae used for liquid biofuel source. Ethanol was generated after lipid extraction	<i>Padina tetrastromatica</i>	Ashokkumar et al., 2017 [159]
12	Improved ethanol was noticed with brown alga using simple pre-processing and drying methods	<i>Laminaria digitata</i>	Adams et al., 2017 [160]
13	Ethanol production was carried out using the enzymatic hydrolysate of brown alga	<i>Saccharina latissima</i>	Lamb et al., 2018 [161]
14	Ethanol production from brown macroalga by an alginate fermentation strain, <i>Meyerozyma guilliermondii</i>	<i>Colpomenia sinuosa</i>	Zhang et al., 2022 [162]

Hou et al. [156] produced ethanol and various proteins using *Laminaria digitata*. Obara et al. [157] used the waste of *Undaria pinnatifida* (kelp), a brown algal species and paper shredder scrap for ethanol production. Both hydrolysates were saccharified with enzymes separately, and the C19 strain of *Saccharomyces cerevisiae* was used for fermentation by mixing both samples (Table 3). Obata et al. [158] worked on the biomass of *Ascophyllum nodosum* and *Laminaria digitata* for ethanol production. They used the non-conventional yeast strains such as *Pichia stipitis* and *Kluyveromyces marxianus* for fermentation. The biomass of *Padina tetrastromatica*, a brown marine macroalga, was used as the source for both ethanol and diesel production [159]. Bioethanol was generated after lipid extraction. After saccharification and fermentation, the ethanol content was 161 mg/g residual biomass. Adams et al. [160] produced a high amount of ethanol from *Laminaria digitata* (brown macroalga) using simple mechanical pre-processing and various drying methods. The enzymatic hydrolysate of *Saccharina latissima* was used for ethanol production through *Saccharomyces cerevisiae*-mediated fermentation [161]. Zhang et al. [162] used the brown alga *Colpomenia sinuosa* for ethanol production. They used the alginate fermentation strain *Meyerozyma guilliermondii* to obtain ethanol (Table 3). Notably, brown algae have been used for the production of various bioproducts, which includes secondary metabolites.

5.3. Production of Ethanol from Red Algae

Some of the red algal species accumulated more carbohydrates, which will be helpful for ethanol production [163]. Among 55 species tested, the acid hydrolysate of *Kappaphycus alvarezii* proved to be the best with respect to ethanol production. Overall, 0.21 g of ethanol was noticed per gram of galactose in this alga (Table 4). The enzymatic hydrolysate of *Gracilaria verrucosa* generated more ethanol through *Saccharomyces cerevisiae*-assisted fermentation [164]. On the whole, high ethanol yield (0.43 g/g sugar) was noticed with the feedstock of this red alga. The ethanol was produced from the biomass of a red alga, namely, *Gelidiella acerosa*, through simultaneous saccharification and fermentation. They used the dilute-acid-pretreated samples for ethanol generation [165]. The biomass of *Pterocladia capillacea* was used for ethanol production through sulfuric acid hydrolysis and detoxification methods. Further, the hydrolysates were used for fermentation by taking

thermo-tolerant yeast *Kluyveromyces marxianus* [166]. Autoclave-treated *Gelidium amansii* (red alga) biomass was used for conversion of sugars into ethanol through SHF and SSF methods [167]. The autoclaved sample with SSF method is superior for ethanol production.

Table 4. Production of ethanol from different red algae.

S.No	Ethanol Production Work	Name of the Algae	Reference
1	The acid hydrolysates of red algae were used for ethanol production	Red algae including <i>Kappaphycus alvarezii</i>	Meinita et al., 2012 [163]
2	Ethanol production was achieved with red alga in a biorefinery approach	<i>Gracilaria verrucosa</i>	Kumar et al., 2013 [164]
3	Ethanol was generated using red alga through simultaneous saccharification and fermentation of dilute-acid-pretreated samples	<i>Gelidiella acerosa</i>	Babujanarthanam and Kavitha, 2014 [165]
4	Ethanol was produced from acid hydrolysis and detoxification methods, and fermentation with thermotolerant <i>Kluyveromyces marxianus</i>	<i>Pterocladia capillacea</i>	Wu et al., 2014 [166]
5	Red algal species was used for ethanol production by autoclaving and SHF and SSF	<i>Gelidium amansii</i>	Kim et al., 2015 [167]
6	Rapid ethanol production was achieved through mutant yeast using biomass of red alga	<i>Gelidium amansii</i>	Lee et al., 2015 [168]
7	Enzymatic hydrolysate of freshwater red alga generated more ethanol with SSF method than seawater red alga	<i>Porphyridium cruentum</i>	Kim et al., 2017 [169]
8	Ethanol was generated from industrial algae waste	<i>Eucheuma spinosum</i>	Alfonsin et al., 2019 [170]
9	Ethanol production from agarophyte red seaweed using a novel sample preparation method for analyzing ethanol content by GC	<i>Gelidium elegans</i>	Hessami et al., 2019 [171]
10	Production of bioethanol from red seaweed by SSF method	<i>Gracilaria verrucosa</i> <i>Eucheuma cottonii</i>	Wadi et al., 2019 [172]
11	Enhanced ethanol production was achieved with hydrolysate of red alga by improvement of catabolite regulatory genes in yeast	<i>Gloiopeltis furcata</i>	Park et al., 2021 [173]
12	Ethanol production was improved using galactose-adapted yeasts with red alga	<i>Eucheuma denticulatum</i>	Kim et al., 2021 [174]

Galactose in *Gelidium amansii* was rapidly converted into ethanol through a mutant *Saccharomyces cerevisiae* HJ7-14 resistant to 2-deoxy-D-glucose [168]. HJ7-14 generated 7.4 g/L ethanol from the hydrolysates of this red alga within 12 h. Both freshwater and sea water *Porphyridium cruentum* (red alga) were tested to determine the ethanol production levels using their enzymatic hydrolysates by applying both SSF and SHF [169]. High ethanol content was produced with freshwater *Porphyridium cruentum* with the SSF method (Table 4). Alfonsin et al. [170] worked on ethanol production using industrial algal waste. They chose the waste of *Eucheuma spinosum* (red alga) and performed acid hydrolysis and fermentation to obtain more ethanol. The biomass of *Gelidium elegans* (red seaweed) was hydrolyzed with 2.5% (*w/v*) H₂SO₄ at 120 °C for 40 min, which yielded more ethanol in fermentation. Ethanol content was estimated by gas chromatography using a novel sample preparation method [171]. The high cellulosic biomass of both *Gracilaria verrucosa* and *Eucheuma cottonii* was used for ethanol production through the simultaneous saccharification and fermentation method [172]. An improvement in ethanol production was achieved with the hydrolysate of *Gloiopeltis furcata* through the enhancement of catabolite regulatory genes in *Saccharomyces cerevisiae* [173]. Kim et al. [174] produced bioethanol using hydrolysates of *Eucheuma denticulatum* by galactose-adapted yeast. Adaptive evolution of microorganisms will be useful for greater ethanol production in the near future. Due to the low carbohydrate content in certain red algal species, less ethanol production work was noticed.

5.4. Production of Ethanol from Blue–Green Algae

Blue-green algae or cyanobacteria are one of the useful groups for the production of bioproducts such as food, cosmetics, nutraceuticals and biofuels. Kampfe [175] worked on ethanol production using the wild algae along with *Spirulina* through batch fermentation (Table 5). A blue-green alga *Microcystis aeruginosa* was used for ethanol production using different media compositions. Also, they used various fermentation strains such as *Saccharomyces cerevisiae*, *Pichia stipitis* and *Zymomonas mobilis* [176]. Sugar content was high in the modified BG11 medium (inclusion of urea along with increased Fe and decreased Ca). *Pichia stipitis* generated more ethanol during fermentation when compared with the other two strains and altogether produced high alcohol content. A popular blue-green alga with more carbohydrates, *Spirulina platensis*, was used for ethanol production by Markou et al. [177]. They used four acids (H_2SO_4 , HNO_3 , HCl and H_3PO_4) with four concentrations (2.5 N, 1 N, 0.5 N and 0.25 N) for saccharification by maintaining four temperatures (40 °C, 60 °C, 80 °C and 100 °C). High yields of ethanol (16.32 and 16.27%) were obtained in both 0.5 N HNO_3 and 0.5 N H_2SO_4 treatments.

Table 5. Production of ethanol from various blue-green algal species.

S.No	Ethanol Production Work	Name of the Algae	Reference
1	Ethanol was produced using wild algae along with blue-green alga	Wild algae along with <i>Spirulina</i>	Kampfe, 2010 [175]
2	Ethanol production was achieved with microalga using different media compositions and <i>Saccharomyces cerevisiae</i> , <i>Pichia stipitis</i> and <i>Zymomonas mobilis</i>	<i>Microcystis aeruginosa</i>	Kimet al., 2012 [176]
3	Blue-green alga was used for ethanol production with four acids for saccharification	<i>Spirulina platensis</i>	Markou et al., 2013 [177]
4	Microalgal species was used for the production of ethanol	<i>Spirulina</i>	Hossain et al., 2015 [178]
5	Blue-green alga was used for ethanol production with various factors along with three fermentation organisms	<i>Microcystis aeruginosa</i>	Khan et al., 2016 [179]
6	CaO pretreatment and combination of four microorganisms for fermentation improved the ethanol production in blue-green alga	<i>Microcystis aeruginosa</i> (KMMCC-1135)	Khan et al., 2017 [180]
7	Augmented ethanol was achieved through optimization of carbohydrate productivity in blue-green alga	<i>Spirulina</i>	Tourang et al., 2019 [181]
8	Ethanol was produced from cyanobacteria-based agro-industrial wastewater treatment and raisin residue extract	<i>Leptolyngbya</i> sps	Tsolcha et al., 2021 [182]
9	Ethanol was generated along with semi-continuous algal–bacterial wastewater treatment	<i>Cyanobacteria</i>	Papadopoulos et al., 2023 [183]

The biomass of *Spirulina* cultures were used for ethanol production through acid hydrolysis and *Saccharomyces cerevisiae*-mediated fermentation [178]. The feedstock of *Microcystis aeruginosa* was used to generate ethanol using different factors. The authors used a modified BG11 medium; various ingredients such as lysine, alanine, aminolevulinic acid and naphthalene acetic acid; and different light conditions [179]. The best growth rate and greater carbohydrate content were observed with red LED light at 25 °C. A high ethanol yield was noticed with the combination of *Saccharomyces cerevisiae*, *Pichia stipitis* and *Brettanomyces custersainus* strains in fermentation (Table 5). To obtain a higher yield of fermentable sugars, *Microcystis aeruginosa* biomass was pretreated with CaO before acid and enzymatic hydrolysis. More ethanol content was noticed in fermentation assisted by the combination of four microorganisms [180]. Tourang et al. [181] augmented the ethanol

production in *Spirulina* by enhancing the carbohydrate production. Tsolcha et al. [182] augmented the ethanol production using cyanobacteria-based biomass in combination with a raisin residue extract as feedstock. They achieved 85.9% of the theoretical ethanol yield. Papadopoulos et al. [183] worked on algal-bacterial wastewater treatment, and harvested biomass was used for ethanol production after acid hydrolysis and fermentation by *Saccharomyces cerevisiae*.

Ethanol production was also achieved from different algal species in one experiment apart from specific species. In addition, some of the research groups used mixed algal species for ethanol production (Table 6). Kim et al. [184] produced ethanol using the hydrolysates of *Ulva lactuca*, *Gelidium amansii*, *Laminaria japonica* and *Sargassum fulvellum*; they used *E. coli* KO11 strain for ethanol production after acid hydrolysis and enzyme treatment of the biomass. Bioethanol production was achieved using *Schizocytrium* species through hydrothermal treatment and biological conversion [185]. Gupta et al. [186] conducted fermentation using the feedstock of marine algal species along with the strain of *Saccharomyces cerevisiae* VITS-M2. Shokrkar et al. [5] proved that enzymatic hydrolysis yielded more ethanol in yeast fermentation at the end than acid hydrolysis with mixed microalgal biomass. Modeling and sensitivity analysis was carried out for ethanol production by enzymatic hydrolysis of microalgal cellulose [187]. High levels of fermentable sugars were obtained with more cellulase (three times); thus, more ethanol was generated in the fermentation. Amomou et al. [188] used both *Ulva lactuca* (green alga) and *Gelidium sesquipedale* (red alga) for ethanol production. For better sugar release and ethanol production, mechano-enzymatic deconstruction was conducted using a new enzyme cocktail, two types of milling modes and different fermentation timings. *Ulva lactuca* generated more ethanol after 72 h of fermentation. Marine macroalgal strains such as *Jaina rubens* (red alga), *Ulva lactuca* (green alga) and *Sargassum latifolium* (brown alga) were used for ethanol production [189].

Table 6. Production of ethanol from different algal species.

S.No	Ethanol Production Work	Name of the Algae	Reference
1	Marine algal hydrolysates were used for ethanol production using <i>Escherichia coli</i> KO11	<i>Ulva lactuca</i> , <i>Gelidium amansii</i> , <i>Laminaria japonica</i> , <i>Sargassum fulvellum</i>	Kim et al., 2011 [184]
2	Hydrothermal treatment and biological conversion were applied for ethanol production	<i>Schizocytrium</i> sps	Kim et al., 2012 [185]
3	Marine algae was used for ethanol production by yeast fermentation	Marine algae	Gupta et al., 2012 [186]
4	Ethanol content was greater with enzymatic hydrolysis than acid hydrolysis	Mixed microalgae	Shokrkar et al., 2017 [5]
5	Modeling and sensitivity analysis were carried out for ethanol production through enzymatic hydrolysis of microalgal cellulose	Mixed microalgae	Shokrkar et al., 2018 [187]
6	Both red and green macroalgal species were used for ethanol production using mechano-enzymatic deconstruction	<i>Ulva lactuca</i> , <i>Gelidium sesquipedale</i>	Amamou et al., 2018 [188]
7	Batch ethanol production through the biological and chemical saccharification of some Egyptian marine macroalgae	<i>Jania rubens</i> , <i>Ulva lactuca</i> , <i>Sargassum latifolium</i>	Soliman et al., 2018 [189]
8	A kinetic model was developed on ethanol production from enzymatic hydrolysates of microalgal biomass	Microalgae	Shokrkar and Ebrahimi, 2019 [190]
9	Ethanol production from defatted biomass of microalgae grown under mixotrophic conditions	<i>Nannochloropsis oculata</i>	Fetyan et al., 2022 [191]

A high content of ethanol production was achieved with algal hydrolysates in a separate batch fermentation process. An enhanced ethanol (0.37 g/g sugar) yield was noticed with *Pachysolen tannophilus*-mediated fermentation. Ethanol production from enzymatic hy-

drollysates of microalgal biomass was carried out by Shokrkar and Ebrahimi [190]; they also used the kinetic models on ethanol production (Table 6). *Nannochloropsis oculata* culture was carried out using agricultural waste and sugarcane bagasse aqueous extract (SBAE) [191]. Augmented ethanol production was observed in defatted biomass grown mixotrophically (SBAE mixotrophic) and pretreated with acid-coupled enzyme hydrolysis. The nitrate limitation in algal culture improved the carbohydrate and starch content. Overall, the information about ethanol-producing algae will be useful for commercial practice in the near future.

6. Future Prospects

Bioethanol is considered to be the best blending fuel with efficient energy content. There are various existing biological resources available for ethanol production apart from chemical resources. Among all the methods, the production of ethanol from biological sources is environmentally friendly with less pollution. Specifically, primitive plants or photosynthetic organisms such as algae can produce several commercial products, including ethanol. However, these algal species are often neglected and considered waste in many places. Hence, identification of novel candidate algal species is a key concern for the production of biofuels, including ethanol. Generally, the rate of ethanol production depends on the efficiency of photosynthesis and biomass. Therefore, it is necessary to focus on the augmentation of biomass and, by extension, the production of useful carbohydrates/sugars for ethanol conversion. A number of physicochemical factors are involved in photosynthesis, biomass accumulation, formation of sugars, hydrolysis of sugars, sugar conversion into ethanol etc. These key steps need to be focused on for sustainable ethanol production from various algae. Figure 5 represents various factors involved in carbohydrate/sugar formation in photosynthesis.

Nutrients are the major factors for photosynthesis apart from water and light conditions. Generally, tris-acetate phosphate medium is universal for most of the in vitro algal cultures. Apart from the TAP medium, BGII medium, Chu medium etc., are well established for both marine and freshwater algal species. Therefore, focusing on the alteration of media components and preparation of nano-sized elements/compounds of various media will be helpful for the improvement of biomass and carbohydrates formation, which, in turn, leads to the enhancement of ethanol production. Recently, we improved the growth and ethanol content in a model alga, *Chlamydomonas reinhardtii*, with the addition of optimal sodium bicarbonate in the TAP medium [11]. In vitro handling of marine algal species requires saline conditions in the culture medium, which is one of the focused issues at present to use more marine species for ethanol production. Growth factors, particularly carbon dioxide, are crucial for photosynthesis or carbohydrate formation apart from water resource. In addition, nitrogen and phosphorous are key factors for the growth of algal species, which need to be focused. Photolysis of water at photosystem II (PSII) is one of the crucial steps in photosynthesis and releases electrons for the electron transport system. Hence, it must include compounds like titanium dioxide (TiO₂) and bismuth vanadate (BiVO₄) in the algal medium, which improve the photolysis of water and, thus, improve the photosynthetic rate.

Vitamins are crucial factors for algal growth and development. Each alga has its own vitamin composition, but the inclusion of additional vitamins augments the biomass, including sugar levels, which is useful for more ethanol generation. Our laboratory achieved the improvement of biomass and ethanol levels with the feedstock of *Chlorococcum minutum* through the inclusion of a vitamin composition (B1, B7 and B12) to the TAP medium [18]. Light, pH ranges and temperature are key factors for algal growth and development, including photosynthesis. Also, these physical factors are crucial in sugar hydrolysis and fermentation processes. Hence, it is necessary to standardize these factors for each alga and strain of fermentation. Our laboratory proved again that dark fermentation yielded more ethanol with *Chlorella vulgaris* feedstocks [13]. Hydrolysis of sugars plays a crucial role in ethanol production. Therefore, a novel pretreatment method is essential for effi-

cient ethanol production. Moreover, identification and isolation of competent hydrolytic enzymes from various biological sources must be focused. It is also essential to develop efficient fermentation strains to augment ethanol production.

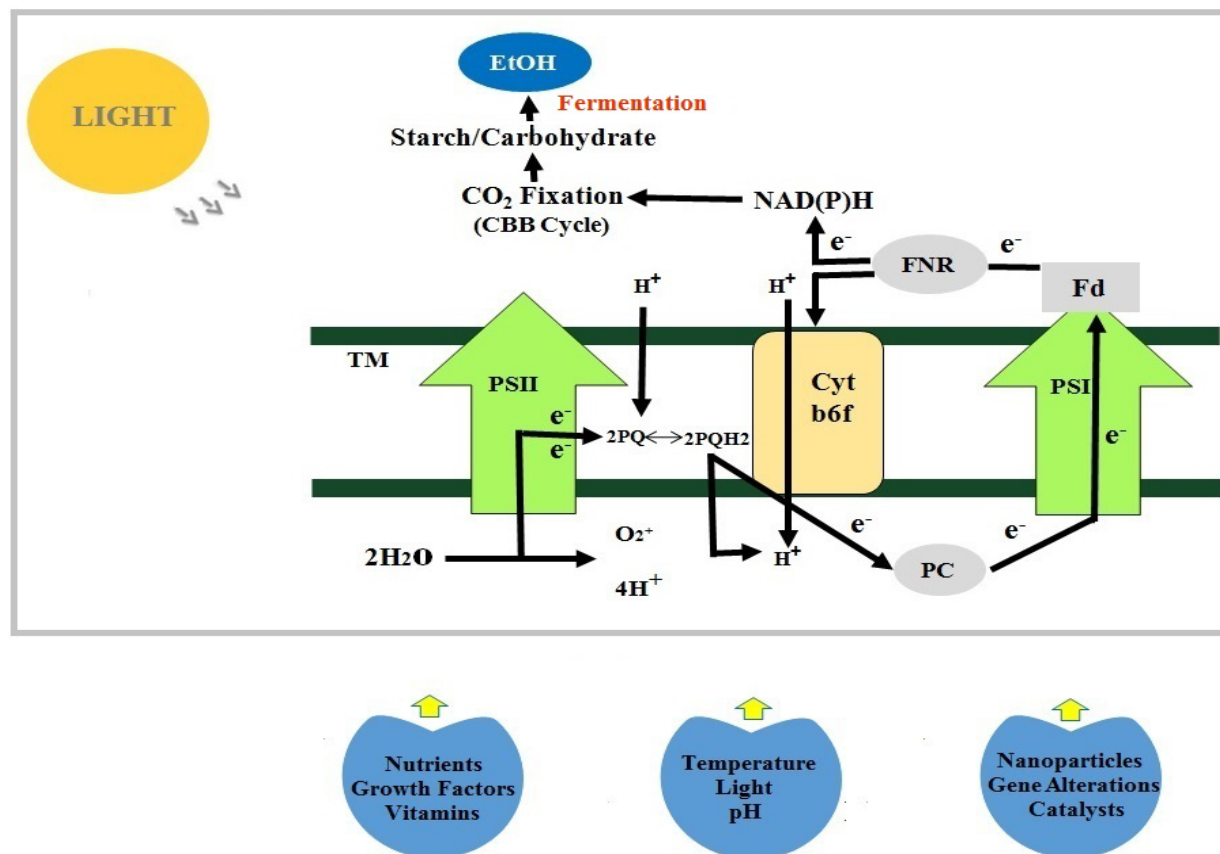


Figure 5. Various factors involved in carbohydrate/sugars formation during photosynthesis in algal cells. CBB cycle—Calvin–Benson–Bassham cycle, Cyt.b6f—Cytochrome b6f, EtOH—Ethanol, Fd—Ferredoxin, FNR—FerredoxinNADP⁺ reductase, NADPH—Nicotinamide adenine dinucleotide phosphate-oxidase, PC—Plastocyanin, PSI—Photosystem I, PSII—Photosystem II, TM—Thylakoid membrane, PQ—Plasto quinone. Boxes below represent different factors involved in photosynthesis and sugar evolution.

Nanoscience is highly useful for photosynthesis; especially several nanoparticles are helpful in various metabolic pathways. In addition, the nano-sized particles are involved in sugar hydrolysis and ethanol formation, which need to be a focus in the near future. Certain nano-sized electrocatalysts (Pt, Pd and Ru), capable electrocatalysts (Pt–Ru, Pt–Pd and Pd–Ru), efficient photocatalysts (TiO₂ and BiVO₄) and RuO₂, ZrO₂, ZnO, CuO etc. will be useful for photosynthesis, sugar hydrolysis and fermentation processes, which are some of the serious concerns in recent times. Moreover, the inclusion of nanocatalysts augmented the chemical reactions, which improved the biomass and ethanol production. Recently, our laboratory identified that Pt–Ru/RGO (both 0.5 and 1.0 mg/L) in the TAP medium yielded more biomass and ethanol contents in *Chlorococcum minutum* and assumed its role in the electron transport system of photosynthesis [24]. Also, it was proved that graphenaceous materials were explored as facile electron-conducting materials for various electrochemical reactions. These kinds of electrocatalysts/photocatalysts may be useful to generate more electrons, which enhance the photosynthesis and sugar levels.

Ethanol production from different algal species was discussed in detail in the above paragraphs, indicating that each and every alga has its own specificity for growth and carbohydrate formation, and thus, ethanol production. Another important factor deter-

mining carbohydrate formation is the genetic background of the algal species. Hence, it is important to focus on gene alterations in algal cells for the improvement of biomass and carbohydrate formation. Moreover, most of the algal species reproduce in an asexual mode and there is no chance of gene escape, which is one of the advantages for genetic engineering research in algal species. Using an advanced genome editing (CRISPR/Cas9) technique in algal cells specifically during the photosynthetic process may result in better carbohydrate generation, which is useful for improving ethanol production. Overall, it is very important to focus on these key aspects to improve the production of carbohydrates and, thus, ethanol from algal species. Apart from implementing these advanced techniques, it is equally important to screen various algae to select the best species with respect to ethanol production. Later, it will be easy to proceed by choosing candidate algal species for ethanol production in vivo instead of working on in vitro experiments. A focus on algal waste management practices is also one of the opportunities for bioethanol production. Furthermore, interdisciplinary research and financial support from various organizations will certainly be useful to generate efficient bioethanol from algal species.

7. Conclusions

At present, the world is facing an energy crisis due to the intense utilization of fossil fuels for industries and motor vehicles as well as domestic purposes. Therefore, it is indispensable to look into alternate renewable energy sources, including bioethanol. Hence, the production of bioethanol is one of the important targets for blending with petrol or diesel, which reduces the cost as well as environmental pollution. In the present review, ethanol production methods and their sources were emphasized in detail. Also, the various factors involved in algal growth and, by extension, ethanol generation were discussed. Moreover, various algal species involved in bioethanol production were highlighted for their commercial use. Further, the inclusion of additional nutrients, growth factors, vitamins, and the alteration of light, pH and temperature may yield better results with respect to the improvement of algal biomass and ethanol production. The addition of various catalysts and nanoparticles involved in different biochemical pathways, including photosynthesis of algal species, can synthesize more carbohydrates/sugars, which are useful for enhancing ethanol production. Gene modifications in algal cells and yeast or bacterial cells may result in better ethanol yield. In conclusion, this review will be useful for industrial production of bioethanol from algae, which is environmentally friendly, and this kind of advanced technology-based market may grow rapidly in the near future.

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